

Understanding the Fundamentals of the Blockchain

Blockchain technology is a constantly evolving, secure and shared record-keeping system in which each user has a copy of the data, which can only be modified if all parties involved in a transaction agree. This article examines the fundamentals of blockchain technology.



To understand the fundamentals of the blockchain, we need to understand how a transaction can be added to the existing blockchain. This can be done by going through a few important steps, as illustrated in Figure 1.

Step 1: Assume that user 1 wishes to transmit to user 2 an X amount of digital bitcoins. The transaction is then initiated by user 1.

Step 2: The node of user 1 initiates a transaction by creating it first and then digitally signing it with its private key. In a blockchain, a transaction can represent a variety of actions.

Step 3: The requested transaction is broadcast to each individual computer via a peer-to-peer (P2P) network (or node).

Step 4: Individual nodes receive the request and use cryptography techniques to try to validate the

transaction. The miner node is the node that completes the validation procedure. Bitcoin is used to compensate these miner nodes.

Step 5: The verified blocks are now added to the blockchain network. Hashing is used to connect these nodes.

Step 6: The approved transactions are then recorded in a public ledger. Transactions are completed and the ledger is updated after the block is added to an existing chain.